

Carlos Andrés Peña Arias

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PROFESSIONAL SUMMARY

Systems & Computing Engineering student focused on security engineering, algorithms, and high-impact problem solving. Known for clear mentorship, disciplined execution, and a strong commitment to quality and integrity.

EDUCATION

Systems and Computing Engineer — Bachelor of Science <i>Universidad de los Andes</i> GPA: 4.57/5.00 Selected ICPC Team Member — Competed at the national and regional levels Research FLAG Lab Member — Post-Quantum Cryptography & Quantum Computing Selected Table Tennis Team Member — Competed at the national level	2022 – 2026 Bogotá, Colombia
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EXPERIENCE

Teaching Assistant — Computer Science & IT <i>Universidad de Los Andes</i> - Provided academic support across Introduction to Programming, Language Theory, and IT in Organizations - Supported 250+ students, earning a 90% average TA rating - Produced reference solutions for 14 Python programming labs through review and rework	2023 – 2025 Bogotá, Colombia
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SKILLS

Programming Languages C++, Python, Java, TypeScript	Core Computer Science Algorithms, Data Structures, Software Design, Computer Networks, Cryptography
Security & Systems Engineering Penetration Testing (PTES), Linux, Wireshark, OpenVAS/Nessus, Nmap, Hashcat	Frameworks & Platforms AWS, GCP, Docker, VMware, Git, React, NestJS, NVIDIA FLARE, CUDA, PyTorch, Power BI
Machine Learning & AI Federated Learning, Graph Neural Networks, Self-Supervised Learning, Differential Privacy, Anomaly Detection	Spoken Languages - English (Full professional proficiency — CEFR: C1) - Spanish (Native)

PROJECTS

Federated Network Intrusion Detection System (FL-NIDS)  <i>Universidad de Los Andes</i> - Designed a federated intrusion detection system to detect malicious network traffic without centralizing sensitive network data - Built the distributed training pipeline using NVIDIA FLARE, leveraging the GraphIDS self-supervised GNN architecture for network flow representation learning - Integrated secure aggregation and differential privacy, running simulations to analyze privacy-accuracy trade-offs in federated training	2026 – Present
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HONORS & AWARDS

Qiskit Fall Fest Hackathon — Universidad de los Andes (IBM-supported) 2nd Place — Quantum computing challenge using Grover’s Algorithm	2025
International Collegiate Programming Contest (ICPC) 15th Place — South America North Regional out of 57 teams Best Team from Universidad de los Andes — National & Regional competitions - Competed in ICPC Colombia and South America North Regionals (2024–2025), demonstrating sustained improvement in algorithmic problem solving and collaborative performance under pressure	2025

CERTIFICATIONS

Google Skills — Google Cloud Essentials 	2026
NVIDIA Deep Learning Institute — Fundamentals of Deep Learning 	2024